

Glintt Next Recruitment Process

Study / Assessment Case

Objective

This case study aims to evaluate three main areas: data engineering skills, data architecture design, and presentation skills.

Candidates are free to use any presentation tool they prefer, as long as it allows exporting to PDF. However, Python must be used for data processing.

Expected Outputs

- A Data Platform Architecture Diagram, including a brief explanation of each component and the rationale behind its selection.
- A diagram of a star schema containing at least one fact table and two dimension tables, along with a brief explanation of the type of Slowly Changing Dimension (SCD) used for each field in the dimension tables.
- An implemented pipeline based on the proposed architecture, with the code published in a public GitHub repository.

Hint: The code should successfully pass a `mypy` type check.

Case

Step 1

Imagine that you have been hired by the International Olympic Committee (IOC) to develop an analytics platform.

SaaS solutions will not be considered valid for this requirement.

- This platform must be capable of supporting batch processing.
- The platform must support not only analytics applications but also machine learning and deep learning workloads.

- It must include a governance layer.

You must justify your architectural choice to the executive board in a text of no more than 20 lines.

Step 2

After designing the requested architecture, the IOC provided a sample dataset and asked you to create a simple Python pipeline to process the data in batches.

For this pipeline, you must:

- Create a diagram of a star schema.
- Specify the type of Slowly Changing Dimension (SCD) used for each column in the dimension tables.
- Implement at least one SCD type in your pipeline.
- The script must follow the same proposed architecture

Data sample:

<https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>